Synoptic Question 23: The Physics of a Maglev Train

Reading Passage

Read the following article and then answer the questions that follow.

Maglev trains, which use magnetic fields for both levitation and propulsion, represent a significant advancement in high-speed ground transport. One type, the "electrodynamic suspension" (EDS) system, uses superconducting magnets (SCMs) mounted on the train. As the train moves at high speed past conductive coils in the guideway, it induces eddy currents in these coils. By Lenz's law, these currents create a repulsive magnetic field, which levitates the train.

This levitation force F_L is negligible at low speeds and increases with speed v. A simplified model for this force is given by:

$$F_L = F_{max}\left(1 - e^{-v/v_c}\right)$$

where F_{max} is the maximum possible lift force and v_c is a characteristic speed. The train must use wheels until it reaches a "lift-off" speed v_{lift} at which F_L equals the train's weight.

Propulsion is achieved by a "linear synchronous motor". The guideway is equipped with electromagnets that are fed an alternating current (AC) of frequency f. This creates a moving magnetic field that "pulls" the SCMs on the train, propelling it forward without contact. The speed of the train v is synchronised with this field and is given by $v = 2df$, where d is the fixed distance between the guideway magnets.

The SCMs must be cooled to liquid helium temperatures (4.2 K) to maintain superconductivity. The energy stored in their powerful magnetic fields is immense and must be managed carefully.

Data:

Data for a prototype Maglev train:

Table 1: Data for Maglev System

Parameter	**Value**
Total mass of train, m	4.50×10^5 kg
Maximum levitation force, F_{max}	6.00×10^6 N
Characteristic speed, v_c	28.0 m s^{-1}
Distance between guideway magnets, d	1.25 m
Total magnetic field energy stored in SCMs, E_{mag}	5.20×10^7 J
Inductance of one SCM, L	25.0 H
Cruising speed of train	500 km h^{-1}
Mass of liquid helium in reservoir	1500 kg
Specific latent heat of vaporization of helium, L_v	2.09×10^4 J kg^{-1}

Questions

(a) The train starts from rest on its wheels.

  (i) Calculate the weight of the train.

  (ii) The train "lifts off" when F_L equals its weight. Calculate the lift-off speed, v_{lift}.

(b) The train is at its cruising speed of 500 km h^{-1}.

  (i) Convert the cruising speed to m s^{-1}.

  (ii) Calculate the frequency f of the AC supply required to maintain this speed.

(c) The superconducting magnets (SCMs) are crucial for both levitation and propulsion.

  (i) Calculate the total current I required to store 5.20×10^7 J of energy in a single SCM of inductance 25.0 H.

  (ii) Explain why it is essential for the SCMs to be superconducting, in terms of power dissipation.

(d) The energy stored in the SCM's magnetic field E_{mag} must be dissipated safely if the magnet "quenches" (loses superconductivity). This energy is dumped into a large resistive load.

(i) If this energy is dissipated over a period of 60 s, calculate the average power dissipated.

(ii) If this same energy were instead used to vaporize the train's liquid helium reservoir at 4.2 K, calculate the mass of helium that would be vaporized.

(e) The levitation force F_L is caused by eddy currents.

(i) State **Faraday's Law of Induction**.

(ii) Use Faraday's Law and Lenz's Law to explain, step-by-step, how the relative motion between the train's SCMs and the guideway coils produces an upward **repulsive** force.

(f) A student wishes to investigate the relationship $F_L = F_{max}(1 - e^{-v/v_c})$.

(i) Rearrange this equation to express $(F_{max} - F_L)$ in terms of v.

(ii) Show how a graph can be plotted from experimental data of F_L and v to determine the constants F_{max} and v_c from a straight line. State the variables to be plotted on the y-axis and x-axis, and how the constants are found.

(g) When the train takes a curved path of radius 3.50 km at its cruising speed, calculate the centripetal acceleration it experiences.

Solutions

(a) (i) $W = mg = (4.50 \times 10^5) \times (9.81) = 4.41 \times 10^6$ N.

(ii) Set $F_L = W$:
$4.41 \times 10^6 = (6.00 \times 10^6)\left(1 - e^{-v_{lift}/28.0}\right)$
$\frac{4.41\times10^6}{6.00\times10^6} = 1 - e^{-v_{lift}/28.0}$
$0.735 = 1 - e^{-v_{lift}/28.0}$
$e^{-v_{lift}/28.0} = 1 - 0.735 = 0.265$
$\ln(e^{-v_{lift}/28.0}) = \ln(0.265)$
$-v_{lift}/28.0 = -1.328$
$v_{lift} = 1.328 \times 28.0 = 37.2$ m s^{-1}.

(b) (i) $v_{cruise} = 500$ km h^{-1} $\times \frac{1000}{3600} = 138.9$ m s^{-1}.

(ii) $v = 2df \implies f = \frac{v}{2d}$
$f = \frac{138.9}{2(1.25)} = 55.6$ Hz.

(c) (i) $E_{mag} = \frac{1}{2}LI^2$
$I = \sqrt{\frac{2E_{mag}}{L}} = \sqrt{\frac{2\times(5.20\times10^7)}{25.0}} = 2040$ A.

(ii) A superconducting material has zero electrical resistance. A normal conductor would have some resistance R. The power dissipated as heat is given by $P = I^2R$. Given the enormous current (2040 A), any non-zero resistance would produce a massive, continuous power loss (heat), making the magnet unfeasible.

(d) (i) $P_{avg} = \frac{E}{t} = \frac{5.20\times10^7}{60} = 8.67 \times 10^5$ W (or 867 kW).

(ii) $Q = mL_v \implies m = \frac{Q}{L_v}$
$m = \frac{5.20\times10^7}{2.09\times10^4} = 2488$ kg.
(This is more than the 1500 kg in the reservoir, so it would vaporize the entire supply).

(e) (i) **Faraday's Law:** The magnitude of the induced e.m.f. in a conductor is directly proportional to the rate of change of magnetic flux linkage through it.

(ii)
- As the train's SCM (a strong magnet) moves, the guideway coils experience a change in magnetic flux.
- By Faraday's Law, this changing flux induces an e.m.f. and thus an eddy current in the coils.
- By Lenz's Law, the direction of this induced current is such that it creates its own magnetic field to **oppose** the change that produced it.
- This "opposition" manifests as a repulsive force between the induced field in the coil and the original field from the SCM, pushing the train upwards.

(f) (i) $F_L = F_{max} - F_{max}e^{-v/v_c}$
$F_L - F_{max} = -F_{max}e^{-v/v_c}$
$F_{max} - F_L = F_{max}e^{-v/v_c}$.

(ii) Take the natural logarithm of the equation from (f)(i):
$\ln(F_{max} - F_L) = \ln(F_{max}e^{-v/v_c})$
$\ln(F_{max} - F_L) = \ln(F_{max}) + \ln(e^{-v/v_c})$
$\ln(F_{max} - F_L) = \left(-\frac{1}{v_c}\right)v + \ln(F_{max})$
This is in the form $Y = mX + c$.
Y-axis: $\ln(F_{max} - F_L)$. (Note: This requires F_{max} to be known. A better way:)
Let's re-read the model: $F_L = F_{max}(1 - e^{-v/v_c})$. We need to find *both* F_{max} and v_c.
Let's check the *rate of change* at $v = 0$. $\frac{dF_L}{dv} = F_{max}(-e^{-v/v_c} \cdot -\frac{1}{v_c}) = \frac{F_{max}}{v_c}e^{-v/v_c}$.
At $v = 0$, the initial slope is $\frac{F_{max}}{v_c}$. This is complex.
Let's try the first method again, assuming F_{max} is found by testing at very high v.
Plot $\ln(F_{max} - F_L)$ on the y-axis against v on the x-axis.
Gradient $= -1/v_c$.
Y-intercept $= \ln(F_{max})$.

(g) $a_c = \frac{v^2}{r} = \frac{(138.9)^2}{3500} = 5.51$ m s^{-2}.

CIE Physics 9702 / Paper 4 (Structured Questions)

Variant 2 (Higher Difficulty): Maglev Train Physics

Instructions (CIE P4 style)

- Answer all questions.
- Use $g = 9.81\,\mathrm{m\,s^{-2}}$.
- Give all working clearly and quote answers to a suitable number of significant figures.

Data for this variant

Parameter	Value
Total mass of train, m	$4.50 \times 10^5\,\mathrm{kg}$
Maximum levitation force, F_{max}	$6.00 \times 10^6\,\mathrm{N}$
Characteristic speed, v_c	$28.0\,\mathrm{m\,s^{-1}}$
Guideway magnet spacing, d	$1.25\,\mathrm{m}$
Cruising speed	$500\,\mathrm{km\,h^{-1}}$
Magnetic energy in SCM system, E_{mag}	$5.20 \times 10^7\,\mathrm{J}$
Curve radius at test section, r	$3.50\,\mathrm{km}$

Question (Variant 2 only)

1 The levitation force is modelled by

$$F_L = F_{\mathrm{max}}\left(1 - e^{-v/v_c}\right).$$

(a) Show that at low speed ($v \ll v_c$), the model predicts an approximately linear relation between F_L and v. State the proportionality constant. [3]

(b) Find the lift-off speed of the train (the speed at which $F_L = mg$). [3]

(c) Open-ended conceptual: Discuss two limitations of using this single-parameter exponential model for real maglev levitation, and for each limitation state one measurable quantity that could be added to improve the model. [4]

2 Propulsion uses a linear synchronous motor with

$$v = 2df.$$

(a) Calculate the AC frequency required for cruising speed $500\,\mathrm{km\,h^{-1}}$. [2]

(b) During acceleration, the train speed rises from $35.0\,\mathrm{m\,s^{-1}}$ to $140\,\mathrm{m\,s^{-1}}$ in $80.0\,\mathrm{s}$. Assuming synchronism is maintained, determine the required rate of change of drive frequency, $\mathrm{d}f/\mathrm{d}t$. [3]

(c) Open-ended conceptual: Explain why loss of synchronism can produce both reduced thrust and thermal stress in guideway coils. Your answer should include magnetic field phase relationship and induced current effects. [4]

3 In a quench event, stored magnetic energy E_{mag} is dumped into a resistor bank.

(a) If this energy is dissipated uniformly in 45.0 s, calculate the mean power. [2]

(b) A safety redesign limits peak dump power to 6.00×10^5 W. Determine the minimum dump time required. [2]

(c) Open-ended design question: Propose and justify two engineering strategies (other than "use a bigger resistor") to reduce quench risk or quench consequences in superconducting magnets. [4]

4 The train travels through a horizontal curve of radius 3.50 km at cruising speed.

(a) Calculate the centripetal acceleration. [2]

(b) Determine the required lateral force on the train. [2]

(c) Open-ended evaluation: The designers claim passenger comfort is acceptable because acceleration is "small". Evaluate this claim using physics and suggest one additional metric (other than acceleration magnitude) relevant to comfort. [4]

[**Total: 35 marks**]

Sample Solutions (Variant 2)

1(a) For $v \ll v_c$, use $e^{-v/v_c} \approx 1 - v/v_c$.

$$F_L \approx F_{\max}\left[1 - \left(1 - \frac{v}{v_c}\right)\right] = \frac{F_{\max}}{v_c} v.$$

Hence $F_L \propto v$ with proportionality constant $F_{\max}/v_c$.

1(b)

$$mg = (4.50 \times 10^5)(9.81) = 4.41 \times 10^6\,\mathrm{N}.$$

Set $F_L = mg$:

$$4.41 \times 10^6 = 6.00 \times 10^6 \left(1 - e^{-v/28.0}\right),$$

$$e^{-v/28.0} = 1 - \frac{4.41}{6.00} = 0.26425,$$

$$v = -28.0 \ln(0.26425) = 37.1\,\mathrm{m\,s^{-1}} \text{ (about } 37\,\mathrm{m\,s^{-1}}).$$

1(c) (example points) Limitations may include neglect of coil heating/resistance change at high induced current, neglect of vertical oscillations and control-loop dynamics, neglect of track irregularities and finite coil geometry, or neglect of magnetic saturation. Improvement variables might include coil temperature, vertical displacement/velocity, guideway geometry factors, or local field strength/current limits.

2(a)

$$v = 500 \times \frac{1000}{3600} = 138.9\,\mathrm{m\,s^{-1}}, \qquad f = \frac{v}{2d} = \frac{138.9}{2(1.25)} = 55.6\,\mathrm{Hz}.$$

2(b) Since $f = v/(2d)$,

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{1}{2d}\frac{\mathrm{d}v}{\mathrm{d}t}.$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{140 - 35.0}{80.0} = 1.3125\,\mathrm{m\,s^{-2}}.$$

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{1.3125}{2(1.25)} = 0.525\,\mathrm{Hz\,s^{-1}}.$$

2(c) (example points) If the travelling field phase speed and train speed differ, magnetic coupling weakens so longitudinal force falls. Phase error also causes larger slip, increasing induced currents in conductive parts and guideway coils; resistive losses I^2R rise, causing heating and thermal stress.

3(a)

$$P_{\text{mean}} = \frac{E}{t} = \frac{5.20 \times 10^7}{45.0} = 1.16 \times 10^6\,\mathrm{W}.$$

3(b)

$$t_{\min} = \frac{E}{P_{\max}} = \frac{5.20 \times 10^7}{6.00 \times 10^5} = 86.7\,\mathrm{s}.$$

3(c) (example points) Valid strategies include active quench detection with fast energy extraction switches, improved cryogenic redundancy and flow monitoring, segmented magnet circuits with bypass diodes, controlled ramp-rate limits, and structural thermal buffering. Credit should depend on clear physics-based justification.

4(a)

$$a_c = \frac{v^2}{r} = \frac{(138.9)^2}{3500} = 5.51\,\mathrm{m\,s^{-2}}.$$

4(b)

$$F_c = ma_c = (4.50 \times 10^5)(5.51) = 2.48 \times 10^6\,\mathrm{N}.$$

4(c) (example points) Since $5.51\,\mathrm{m\,s^{-2}} \approx 0.56g$, "small" is arguable; comfort depends on duration and direction relative to passengers. Another useful metric is jerk (rate of change of acceleration), or unbalanced lateral acceleration after banking/tilt compensation.

Mark Scheme (CIE P4 style, Variant 2)

Question 1

- 1(a) B1 for valid low-x expansion of exponential; M1 for substitution into model; A1 for $F_L \approx (F_{\max}/v_c)v$. [3]
- 1(b) M1 for mg; M1 for correct rearrangement/log step; A1 for $v \approx 37\,\mathrm{m\,s^{-1}}$. [3]
- 1(c) Any two justified limitations (1 mark each) plus matched measurable improvement variable (1 mark each). Max 4. [4]

Question 2

- 2(a) M1 for speed conversion or direct consistent substitution; A1 for $f \approx 55.6\,\mathrm{Hz}$. [2]
- 2(b) M1 for linking f to v and gradient/rate; M1 for acceleration value; A1 for $\mathrm{d}f/\mathrm{d}t \approx 0.525\,\mathrm{Hz\,s^{-1}}$. [3]
- 2(c) Up to 4 marks: phase relation and synchronism (1), reduced thrust mechanism (1), induced/slip current increase (1), thermal consequence via I^2R or equivalent (1). [4]

Question 3

- 3(a) M1 for $P = E/t$; A1 for $1.16 \times 10^6\,\mathrm{W}$. [2]
- 3(b) M1 for $t = E/P_{\max}$; A1 for $86.7\,\mathrm{s}$ (or 87 s). [2]
- 3(c) 1 mark per valid strategy (max 2) and 1 mark per sound physics justification (max 2). [4]

Question 4

- 4(a) M1 for $a = v^2/r$; A1 for $5.51\,\mathrm{m\,s^{-2}}$. [2]
- 4(b) M1 for $F = ma$; A1 for $2.48 \times 10^6\,\mathrm{N}$. [2]
- 4(c) Up to 4 marks: quantitative comfort critique using computed acceleration (1–2), valid interpretation/context (1), suitable additional metric such as jerk or lateral unbalance with explanation (1). [4]

Total: 35 marks